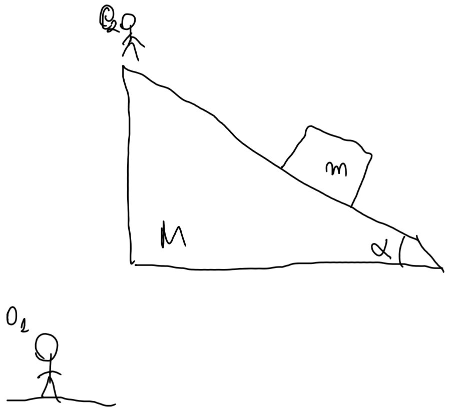


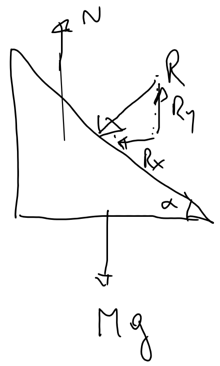
Block is setup on top of a moving wedge.

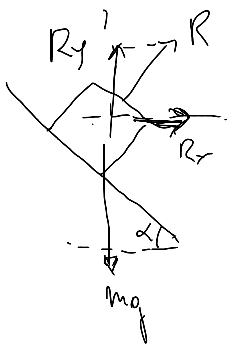
- block → mass m
- wedge → mass M
- no friction between m and M
- no friction between M and ground
- wedge hipotenuse is L

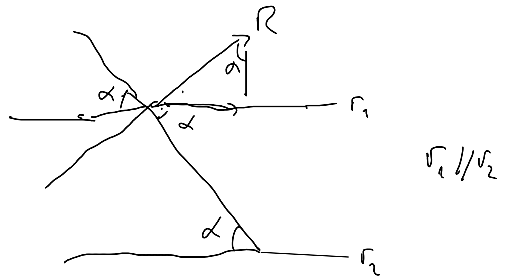
O_1 : observer in the ground at rest, with set of coordinates (x, y)

O_2 : NON-newtonian observer in an accelerated frame with coord. (x', y')

Free Body Diagrams:







(1) $N + R_y - Mg = MA_y$

(2) $-R_x = MA_x$

(3) $R_x = m a_x$

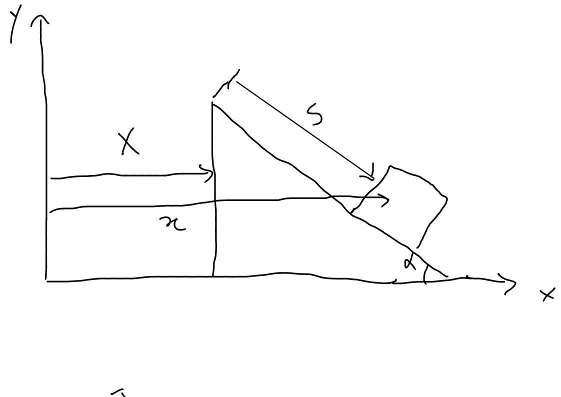
(4) $R_y - mg = m a_y$

(5) $R_x = R \sin\alpha$

(6) $R_y = R\cos\alpha$

Constraints:

(7) $A_y = 0$ → wedge does not move upwards or downwards

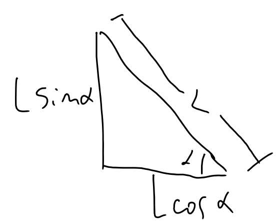


$s = s(t)$ → position of the block as viewed by observer O_2, gives the relative position between block and wedge

$X = X(t)$ → position of wedge with respect to the ground, measured by observer O_1.

ground
observer O_1.

$x = x(t) \rightarrow$ position of block with respect to the ground observer O_1.



(8) $x(t) = X(t) + s(t)\cos\alpha$

(9) $y(t) = L\sin\alpha - s(t)\sin\alpha$

For now we have the following unknowns:

$A_x,\ a_x,\ a_y,\ x(t),\ X(t),\ s(t),\ N,\ R$

Deriving with respect to time (8) and (9)

(10) $\dot{x} = \dot{X} + \dot{s}\cos\alpha$

(11) $\dot{y} = -\dot{s}\sin\alpha$

(12) $\ddot{x} = \ddot{X} + \ddot{s}\cos\alpha \Rightarrow a_x = A_x + \ddot{s}\cos\alpha$

(13) $\ddot{y} = -\ddot{s}\sin\alpha \rightarrow a_y = -\ddot{s}\sin\alpha$

- Notice that $\ddot{s}$ is the relative acceleration between the block and the wedge, as perceived by the observer O_2.

From (2) and (3):

$R_x = MA_x = ma_x \Rightarrow M\ddot{X} = m\ddot{x} \Rightarrow M\dot{X} = m\dot{x} + \text{constant}$

and this gives conservation of momentum, considering $\dot{x}(t=0)=0$, $\dot{X}(t=0)=0$, so constant $=0$, then

(14) $m\dot{x} - M\dot{X} = 0 \therefore \vec{P}_{Before} = \vec{P}_{After}$

Conservation of energy states that:

Conservation of energy states that:

$$E_{BEFORE} = E_{AFTER}$$

$$U_B + K_B = U_A + K_A$$

Before: we only have potential energy, considering the bloce was released from rest from the top of the wedge, at $h(t=0) = L\sin\alpha$.

$$K_B = 0, \quad U_B = mgL\sin\alpha$$

$$K_A = \frac{1}{2}M\dot{X}^2 + \frac{1}{2}m\dot{x}^2, \quad U_A = mg\left(L\sin\alpha - s\sin\alpha\right)$$

$$(15) \quad mgL\sin\alpha = \frac{1}{2}M\dot{X}^2 + \frac{1}{2}m\dot{x}^2 + mg(L-s)\sin\alpha$$

$$\hookrightarrow \quad M\dot{X}^2 + m\dot{x}^2 = 2mgs\sin\alpha, \quad \text{but } M\dot{X} = m\dot{x}$$

$$\hookrightarrow \quad M\left(\frac{m\dot{x}}{M}\right)^2 + m\dot{x}^2 = 2mgs\sin\alpha$$

$$\hookrightarrow (16) \left(\frac{M+m}{M}\right)\dot{x}^2 = 2gs\sin\alpha$$

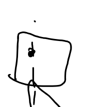
From (12) and (13)

$$\ddot{s} = -\frac{a_y}{\sin\alpha} = \frac{a_x - A_x}{\cos\alpha}$$

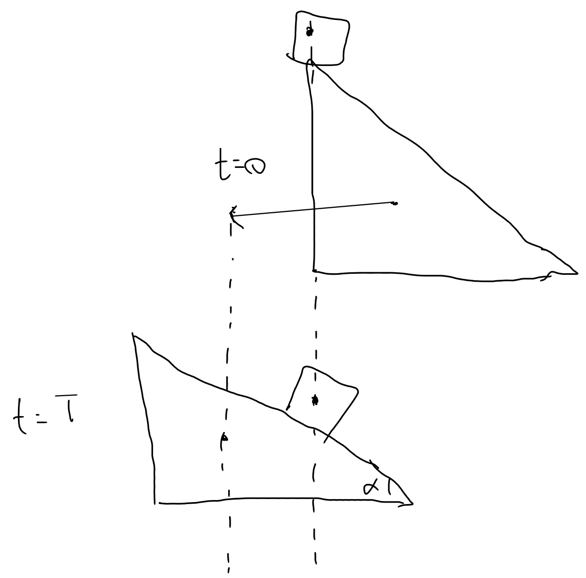
$$\hookrightarrow (17) \quad a_y = (A_y - a_x)\operatorname{tg}\alpha$$

Notice that if $a_y = g$, then the block is in free fall if

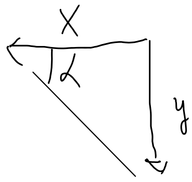
$$(18) \quad A_x = a_x + g\operatorname{cotg}\alpha$$



If we impose that the center of



If we impose that the center of mass of the block does not move in the x-direction, then $a_x = 0$, $\dot{x} = 0$, and in a time $t = T$, the wedge moved $X(T) = AT^2/2$, and the block ~~felt~~ fell by $y = a_y T^2/2 = gT^2/2$



$\therefore \; a_y = (A_y - a_x)\,\mathrm{tg}\,\alpha$

$\therefore \; g = A\,\mathrm{tg}\,\alpha$

$$(19) \quad A_x = g\,\mathrm{cotg}\,\alpha \qquad \text{(for free fall of the block)}$$

Summary:

$$R \sin\alpha = m a_x = -M A_x \Rightarrow R = -m a_x / \sin\alpha$$

$$a_y = (A_x - a_x)\,\mathrm{tg}\,\alpha = \left(\frac{m}{M} - 1\right) a_x\,\mathrm{tg}\,\alpha$$

$$R\cos\alpha - m a_y = mg$$

$$\hookrightarrow \; m a_x \frac{\cos\alpha}{\sin\alpha} - m\left(-\frac{m}{M} - 1\right) a_x\,\mathrm{tg}\,\alpha = mg$$

$$\hookrightarrow \; M a_x \cos^2\alpha + (m + M)\, a_x \sin^2\alpha = gM \sin\alpha\cos\alpha$$

$$\hookrightarrow \; (M\cos^2\alpha + M\sin^2\alpha + m\sin^2\alpha)\, a_x = Mg\sin\alpha\cos\alpha$$

$$\boxed{a_x = \frac{Mg\sin\alpha\cos\alpha}{M + m\sin^2\alpha}}$$

$$\boxed{A_x = -\frac{m}{M} a_x = -\frac{mg\sin\alpha\cos\alpha}{M + m\sin^2\alpha}}$$

$$a_y = (A_x - a_x)\,tg\,\alpha = -\frac{(M+m)\,g\sin\alpha\cos\alpha\,tg\,\alpha}{M + m\sin^2\alpha}$$

$$\boxed{a_y = -\frac{(M+m)\,g\sin^2\alpha}{M + m\sin^2\alpha}}$$

<u>sanity check</u> :

if $M = 0 \Rightarrow A_x = -g\frac{\cos\alpha}{\sin\alpha}$, $a_x = 0$, $a_y = -g$

if $M \gg m \Rightarrow A_x = 0$, $a_x = g\sin\alpha\cos\alpha$, $a_y = -g\sin^2\alpha$

$$\boxed{\ddot{s} = a_{B/W} = -\frac{a_y}{\sin\alpha} = \frac{(M+m)\,g\sin\alpha}{M + m\sin^2\alpha}}$$

$$\frac{a_{B/W}}{A} = \frac{(M+m)\,g\sin\alpha}{M + m\sin^2\alpha} \times \frac{M + m\sin^2\alpha}{mg\sin\alpha\cos\alpha} = \frac{M+m}{m\cos\alpha}$$

$$\boxed{\frac{A}{a_{B/W}} = \frac{m}{M+m}\cos\alpha}$$